Week 9 Reference Sheet: While Loops & Advanced Debugging

CSC-125-201 | Use this as a reference when completing Homework 9

Homework Topics

- Basic while loop syntax
- While loop with counters
- Infinite loop prevention
- break and continue
- Input validation pattern
- Sentinel values
- Common error types
- Reading tracebacks
- try/except basics
- Print debugging

Basic While Loop

Repeats code as long as the condition is True. The condition is checked **only at the top**, not during the body.

```
count = 1
while count ≤ 5:
    print(count)
    count = count + 1  # Must update to avoid infinite loop
```

For vs While

Use `for` when	Use `while` when
You have a list or sequence	You do not know how many repetitions
You know the count: `range(10)`	You are waiting for a condition to change
Processing each item	User input, searching, games

Condition Phrasing

The condition describes when to **keep going**, not when to stop.

You want	Wrong	Right
Stop at 5	`while count == 5`	`while count < 5`
Stop when empty	`while len(items) == 0`	`while len(items) > 0`
Stop on "quit"	`while answer == "quit"`	`while answer ≠ "quit"`

break and continue

Keyword	Effect	Use when
`break`	Exit the loop entirely	Found what you need, user typed "quit"
`continue`	Skip rest of current iteration	Skip invalid data, move to next item

With `continue` in a while loop, update the loop variable **before** the continue to avoid infinite loops.

Input Validation Pattern

```
valid = False
while not valid:
    response = get_input()  # Get user input
    if meets_requirements(response):
        valid = True
    else:
        print("Invalid. Try again.")
```

Sentinel Value Pattern

```
value = get_first_input()
while value ≠ -1:        # -1 is the sentinel
    process(value)
    value = get_next_input()
```

Common Error Types

Error	Cause	Example
`SyntaxError`	Invalid Python syntax	Missing colon, bad indentation
`NameError`	Variable not defined	Typo in variable name
`TypeError`	Wrong type for operation	`"age: " + 25`
`ValueError`	Wrong value for operation	`int("abc")`
`IndexError`	Index out of range	`list[5]` on a 5-item list
`ZeroDivisionError`	Dividing by zero	`10 / 0`

Reading Tracebacks (Bottom-Up)

1. Last line: error type and description. 2. Line above: the code that failed. 3. Above that: file and line number.

try/except

```python
# Catch a specific error
try:
    number = int(user_input)
except ValueError:
    print("Not a valid number.")

# Catch multiple errors
try:
    result = int(val) / int(other)
except ValueError:
    print("Not a number.")
except ZeroDivisionError:
    print("Cannot divide by zero.")

# Catch any error (debugging only)
try:
    risky_code()
except Exception as e:
    print(f"Error: {e}")
```

5-Step Debugging Process

1. **Reproduce** the error. 2. **Read** the traceback bottom-up. 3. **Hypothesize** what went wrong. 4. **Test** with a print statement. 5. **Fix** one thing, then re-run.

Common While Loop Errors

Error	Cause	Fix
Infinite loop	Loop variable never changes	Add update line inside the loop
Off-by-one	`≤` instead of `<`	Trace the last iteration by hand
Never enters loop	Condition is False from the start	Check initial variable value
Infinite with continue	Update line is after continue	Move update before the continue

Resources

- W3Schools - Python While Loops
- LearnPython.org - Interactive Loop Tutorial
- Real Python - While Loops Guide
- W3Schools - Try/Except
- Real Python - Understanding Tracebacks